

Chapter 7 — Class 1

Externalities: private decisions and social costs

Instructor preparation

Length: 55 minutes. **Slides:** 1–15. **Reading:** Chapter 7 notes, “An Introduction” through “Exercise — Identifying and Correcting Externalities.”

Learning goals: students can identify a third party, distinguish private from social marginal values, calculate private and efficient quantities, and explain how a corrective tax or subsidy changes incentives.

Board setup: reserve the left side for definitions, the middle for the negative-externality graph, and the right side for calculations. Erase the middle and right for the positive-externality example. Keep these two rules visible:

Private market: $MPB = MPC$ Efficient quantity: $MSB = MSC$.

Say: “Marginal means the effect of one more unit.” Define each abbreviation before using it. The examples assume competitive markets and constant spillovers per unit. Government administration and enforcement costs are omitted.

0–5 minutes | Opening: who is missing? — slides 1–3

Say: “A buyer and a seller may both gain from a trade. A third person may still be harmed. We need to count that person’s cost.”

Use a refinery selling fuel. Ask: “Who buys? Who sells? Who may experience a cost without taking part in the sale?”

Expected answer: fuel buyer, refinery, nearby residents affected by pollution. Take two responses. Separate residents’ pollution damage from the buyer’s spending on fuel.

Then show slide 3. Ask for a quick vote on whether zero pollution is always the best economic target. Do not settle the question yet.

Bridge: “We will compare the benefit and cost of one more unit. This gives us a way to choose a quantity.”

5–12 minutes | The language of externalities — slides 4–9

Write and say:

- Private cost: a cost considered by the participant.

- External cost: a cost imposed on another person and omitted from the participant's incentives.
- Social cost: private cost plus external cost.

$$MSC = MPC + MEC, \quad MSB = MPB + MEB.$$

Read the first equation aloud: "Marginal social cost equals marginal private cost plus marginal external cost." Do the same for benefits.

Say: "Market failure means the market does not maximize total net benefits. A firm can earn a profit in a market that is inefficient."

Give students 30 seconds to identify the third party in each example:

Activity	Private effect	External effect
Driving at rush hour	Driver's own time and fuel	Extra delay imposed on other drivers
Education	Student's skills and earnings	Knowledge that benefits other people

Ask: "Is a student's higher salary itself an external benefit?" **Answer:** no; it is a benefit received by the participant.

Use slides 8 and 9 as quick checks. Expected answers: smoking when others experience smoke; college education when it generates benefits for others. Require a named third party. The activity label alone is not the explanation.

12–27 minutes | Board example 1: the refinery — slide 10

Allow about 3 minutes for setup and the private equilibrium, 5 for the social optimum, 4 for the graph, and 3 for interpretation and responses.

Write: "Illustrative market; quantity in units; prices and marginal values in dollars per unit."

$$D : P = 14 - Q, \quad S : P = 2 + Q, \quad MEC = 4.$$

There is no external consumption benefit, so demand is both private and social marginal benefit.

Step 1: find the private market outcome

Ask: "Which two curves guide buyers and sellers?" Then solve:

$$14 - Q = 2 + Q \Rightarrow 12 = 2Q \Rightarrow Q_M = 6,$$

$$P_M = 14 - 6 = 8.$$

Say: "At six units, the value to the buyer equals the private cost to the seller. The market calculation leaves out pollution damage."

Step 2: count the external cost

$$MSC = (2 + Q) + 4 = 6 + Q.$$

Ask students to predict whether the efficient quantity will be above or below six before solving:

$$14 - Q = 6 + Q \Rightarrow 8 = 2Q \Rightarrow Q_S = 4.$$

Say: “Including the damage raises the cost of each unit. Four units maximize total net benefits in this example.”

Step 3: draw the graph

1. Label the horizontal axis “Quantity” and the vertical axis “Dollars per unit.”
2. Draw downward demand using points (0, 14) and (6, 8); label it $D = MPB = MSB$.
3. Draw upward private supply using (0, 2) and (6, 8); label it $S = MPC$.
4. Draw parallel social marginal cost using (0, 6) and (4, 10); label it MSC .
5. Mark private equilibrium (6, 8) and the social intersection (4, 10). Drop lines to quantities 6 and 4.
6. Draw a vertical bracket of 4 between the two cost curves at the same quantity. Label it “external cost per unit.”

Do not call 10 the unregulated market price. It is the marginal social value and cost at the efficient quantity. It becomes the buyer price under the tax introduced next.

Step 4: explain the excess output

Ask: “At $Q = 5$, what is the benefit? What is the full cost?”

$$MSB(5) = 9, \quad MPC(5) = 7, \quad MSC(5) = 11.$$

Say: “The buyer values this unit at nine dollars. The seller’s private marginal cost is seven. But the full marginal cost is eleven. Private incentives support output that costs society more than it is worth.”

“Too much production” means more than the efficient quantity. All six units may sell; this is not an unsold surplus.

27–37 minutes | Corrective tax and welfare — slides 11–13

Ask: “How could we make the seller face the omitted four-dollar cost?”

Write a tax of $t = 4$ per unit. In terms of the buyer price, taxed supply is:

$$P_B = 2 + Q + 4 = 6 + Q.$$

Demand meets taxed supply at $Q = 4$. Then calculate both prices:

$$\begin{aligned} P_B &= 14 - 4 = 10, & P_S &= 2 + 4 = 6, \\ P_B - P_S &= 4, & \text{tax revenue} &= 4 \times 4 = 16. \end{aligned}$$

Say: “Buyers pay ten. Sellers keep six after the tax. The four-dollar difference is paid to government.”

Ask: “Does charging the seller mean only the seller bears the burden?” **Answer:** no. Relative to the original price of 8, buyers pay 2 more and sellers receive 2 less in this example.

Shade the triangle between demand and MSC from $Q = 4$ to $Q = 6$. Its vertices are $(4, 10)$, $(6, 8)$, and $(6, 12)$.

$$DWL = \frac{1}{2}(6 - 4)(12 - 8) = 4.$$

Say: “This area is the net loss from excess output. The tax removes that loss in our model. The sixteen dollars of revenue is a transfer, not the welfare gain.”

Remaining damage is $4 \times 4 = 16$. Efficient output does not require zero damage here. The resource benefits from the first four units justify their full costs.

Use slide 13: subsidizing production of the harmful good generally moves output in the wrong direction in this model. Regulation can also reduce output, but its design and costs matter.

Quick check: “Does simply knowing about the damage guarantee internalization?” **Answer:** no. Incentives must account for the damage.

37–49 minutes | Board example 2: a positive spillover — slides 14–15

Say: “Now suppose an activity benefits other people. The buyer does not count the whole benefit. The market may provide too little.”

Use education or another service with an assumed external benefit. Keep the numbers from the notes:

$$MPB = 10 - Q, \quad MPC = MSC = 2 + Q, \quad MEB = 4.$$

Give students one minute to calculate the private quantity, then work through the result:

$$10 - Q = 2 + Q \Rightarrow Q_M = 4, \quad P_M = 6.$$

Include the benefit to others:

$$MSB = 14 - Q, \quad 14 - Q = 2 + Q \Rightarrow Q_S = 6.$$

Draw: downward MPB with intercept 10; downward MSB with intercept 14; upward MSC with intercept 2. Mark private intersection $(4, 6)$ and social intersection $(6, 8)$. Mark the vertical benefit gap of 4. Ask students which quantity is larger and why.

At $Q = 5$, private benefit is 5, social benefit is 9, and cost is 7. Say: “This additional output creates a social gain. Private willingness to pay is too low to support it without a change in incentives.”

A subsidy of $s = 4$ creates a difference between the seller’s receipt and the buyer’s payment:

$$P_S = 2 + 6 = 8, \quad P_B = 10 - 6 = 4, \\ P_S - P_B = 4, \quad \text{government spending} = 4 \times 6 = 24.$$

Say: “The seller receives eight dollars per unit. The buyer pays four. Government pays the other four. The external benefit existed before the subsidy.”

If students are keeping pace, shade the missed-gain triangle with vertices $(4, 6)$, $(4, 10)$, and $(6, 8)$; its area is 4. Otherwise state the direction of the welfare effect and keep the exit check on time. Government spending of 24 is not a social-surplus gain of 24.

49–55 minutes | Exit check and transition

Allow three minutes of individual work and three minutes to review.

Prompt: Demand is $18 - Q$, private supply is $2 + Q$, and marginal external cost is 4.

1. Find the private quantity and efficient quantity.
2. Find the corrective tax and the two prices after the tax.
3. Finish: “The private quantity is too high because ...”

Answer key: $Q_M = 8$, $P_M = 10$; $MSC = 6 + Q$; $Q_S = 6$; tax 4; buyers pay 12, sellers receive 8. The market omits costs imposed on third parties. Optional check: initial DWL is $\frac{1}{2}(8 - 6)4 = 4$.

Closing sentence: “Next time we will ask when people can bargain over these effects and how access to a good changes incentives.”

Pacing and optional depth

If behind: omit tax incidence and the subsidy triangle. Keep both quantity calculations, the tax price wedge, and the exit check. If algebra is slow, complete only the negative-externality calculation and explain the positive case with the graph.

Optional extension, outside the plan: verify the tax’s four-dollar welfare gain by accounting for all components. Before tax: consumer surplus 18, producer surplus 18, damage 24, net surplus 12. After tax: consumer surplus 8, producer surplus 8, revenue 16, damage 16, net surplus 16. Revenue is counted once; damage is subtracted once.

Misconceptions to listen for: social means government; all personal benefits are external; over-production means unsold goods; tax revenue is deadweight loss; social efficiency means no remaining pollution.